

DSA0601 Data Handling and Visualization

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SET 1 — MONTHLY SALES DATA

Dataset

Month	Sales (\$)
January	15000
February	18000
March	22000
April	20000
May	23000

PROGRAM 1 — Line Chart

```
# Dataset
```

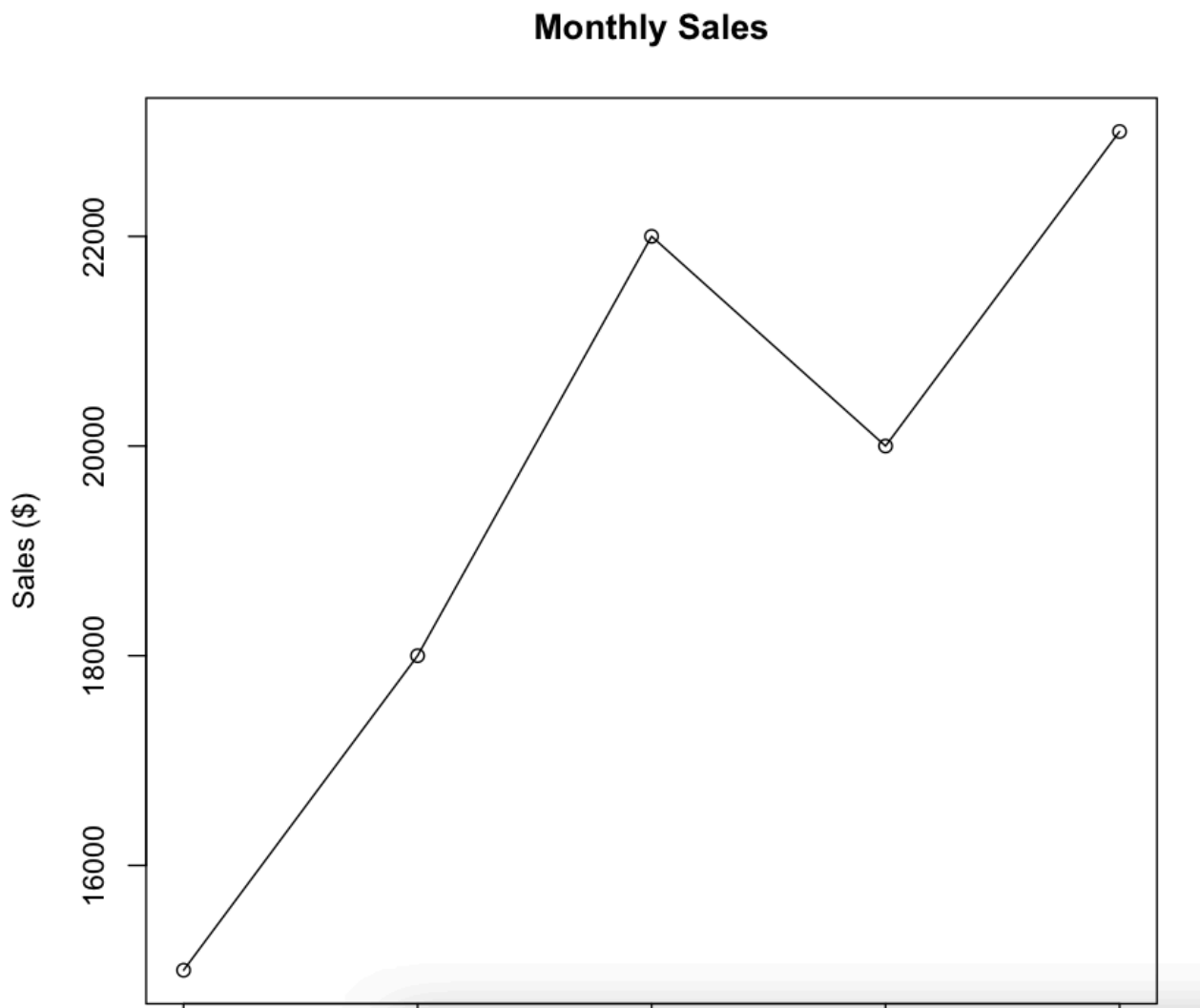
```
month <- c("January", "February", "March", "April", "May")
```

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
# Line Chart
```

```
plot(sales,  
     type = "o",  
     xaxt = "n",  
     xlab = "Month",  
     ylab = "Sales ($)",  
     main = "Monthly Sales")  
axis(1, at = 1:5, labels = month)
```

Output: Line chart showing monthly sales from January to May.



Observation: Sales increase from January to March, decrease slightly in April, and reach the highest value in May.

PROGRAM 2 — Bar Chart

Dataset

```
month <- c("January", "February", "March", "April", "May")
```

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

Bar Chart

```
barplot(sales,  
        names.arg = month,  
        xlab = "Month",  
        ylab = "Sales ($)")
```

```
main = "Monthly Sales")
```

Output: Bar chart comparing sales for each month.



Observation: May has the highest sales of \$23,000, while January has the lowest sales of \$15,000.

PROGRAM 3 — Scatter Plot

```
# Dataset
```

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

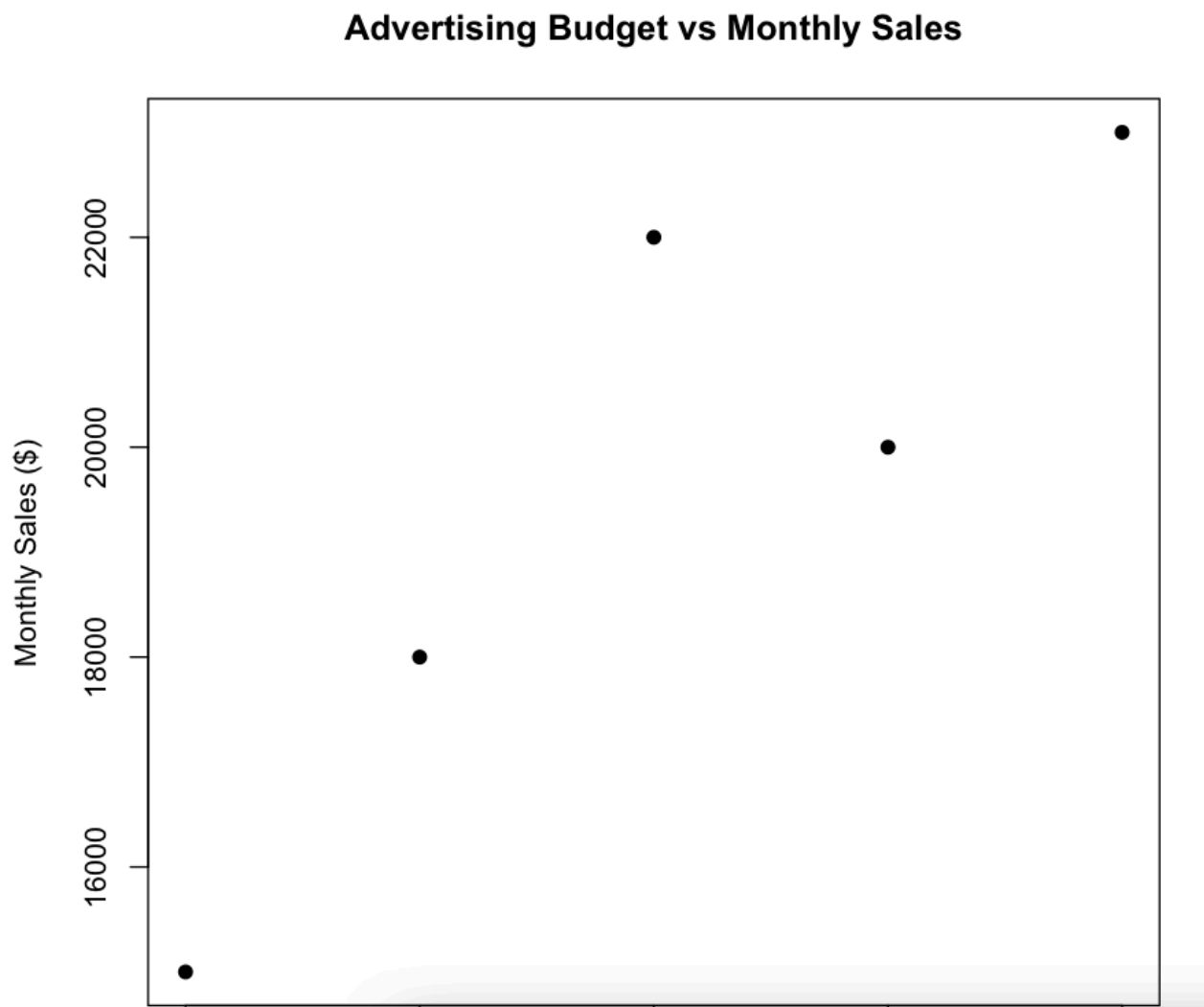
```
advertising <- c(5000, 6000, 7000, 8000, 9000)
```

```
# Scatter Plot
```

```
plot(advertising, sales,
```

```
main = "Advertising Budget vs Monthly Sales",  
xlab = "Advertising Budget ()", ylab = "MonthlySales()",  
pch = 19)
```

Output: Scatter plot showing the relationship between advertising budget and monthly sales.



Insight: The plot shows a generally positive relationship between advertising budget and sales. As the advertising budget increases, monthly sales generally increase.

PROGRAM 4 — Interactive Dashboard using Shiny

```
library(shiny)
```

```
# Dataset
```

```
month <- c("January", "February", "March", "April", "May")
```

```

sales <- c(15000, 18000, 22000, 20000, 23000)
# User Interface
ui <- fluidPage(
  titlePanel("Monthly Sales Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput(
        "chart",
        "Select Chart:",
        choices = c("Line Chart", "Bar Chart")
      )
    ),
    mainPanel(
      plotOutput("salesPlot")
    )
  )
)
# Server
server <- function(input, output) {
  output$salesPlot <- renderPlot({
    if (input$chart == "Line Chart") {
      plot(sales,
        type = "o",
        xaxt = "n",
        xlab = "Month",
        ylab = "Sales ($)",
        main = "Monthly Sales")
      axis(1, at = 1:5, labels = month)
    } else {
      barplot(sales,
        names.arg = month,
        xlab = "Month",
        ylab = "Sales ($)",
        main = "Monthly Sales")
    }
  })
}

```

```
# Run Application
shinyApp(ui = ui, server = server)
```

Output: An interactive dashboard appears with a “**Select Chart**” option. Selecting **Line**

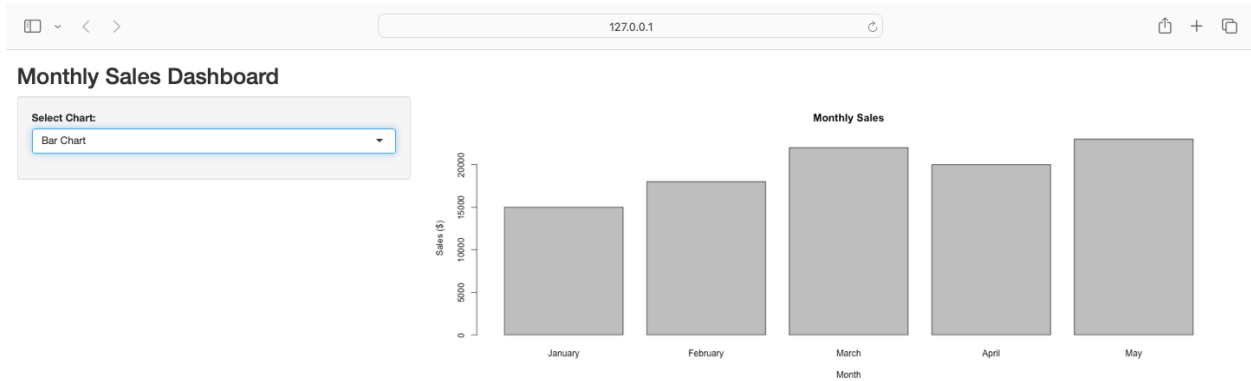


Chart displays the monthly sales trend, while selecting **Bar Chart** displays the monthly sales comparison.

PROCEDURE

1. Create the month and sales vectors containing the monthly sales data.
2. Use the `plot()` function to create a line chart showing the sales trend.
3. Use the `barplot()` function to compare monthly sales.
4. Create advertising and sales vectors and use `plot()` to generate a scatter plot.
5. Load the **Shiny** package and create the user interface and server.
6. Run the Shiny application and switch between the Line Chart and Bar Chart options.

RESULT

Monthly sales data was successfully visualized using **line chart, bar chart, and scatter plot**. An **interactive Shiny dashboard** was also created to display the sales charts.

KEY R FUNCTIONS USED

- `plot()` – Creates line and scatter plots.
- `axis()` – Adds custom month labels to the x-axis.
- `barplot()` – Creates bar charts.
- `fluidPage()` – Creates the Shiny user interface.
- `selectInput()` – Provides chart selection.
- `renderPlot()` – Displays plots dynamically.
- `shinyApp()` – Runs the Shiny application.

SET 2 — CUSTOMER FEEDBACK ANALYSIS

Dataset

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

PROGRAM 1 — Histogram

```
# Dataset
age <- c(25, 30, 35, 28, 40)
# Histogram
hist(age,
      main = "Distribution of Customer Ages",
      xlab = "Age",
      ylab = "Frequency")
```